Hypothesis testing with multiple regression

Recap: hypothesis tests and nested models

Data: Height, age, and gender for 198 children between the ages of 8 and 18. Researchers think the relationship between age and height might depend on gender.

Research question: Is there *any* relationship between Age and Height, after accounting for gender?

Full model:

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Reduced model:

$$\text{Height} = \beta_0 + \beta_1 \text{IsMale} + \varepsilon$$

Recap: hypothesis tests and nested models

Data: Height, age, and gender for 198 children between the ages of 8 and 18. Researchers think the relationship between age and height might depend on gender.

Research question: Is the slope on Age the same for both boys and girls?

Full model:

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Reduced model:

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \varepsilon$$

Nested F test

To compare *any* two **nested** models:

Test statistic: $F = \dfrac{\frac{1}{\#\text{ parameters tested}}(SSE_{reduced} - SSE_{full})}{\frac{1}{n-p} SSE_{full}}$

Distribution: $F_{\#\text{ parameters tested},\, n-p}$

- $n =$ number of observations in data
- $p =$ number of parameters in full model

Intuition: $SSE_{reduced} - SSE_{full}$ is large when the full model fits the data better than the reduced model

Class activity

Work on the class activity (handout).

$SSE_{reduced} = 6733.9$

$SSE_{full} = 1686.3$

$n = 198$

$p = 4$ $\Rightarrow n-p = 194$

Class activity

```
anova(m1)
```

parameters tested = 2

$= df_{reduced} - df_{full}$

$196 - 194 = 2$

```
Response: Height
           Df Sum Sq Mean Sq F value    Pr(>F)
Age         1 5005.2  5005.2 575.820 < 2.2e-16 ***
Sex         1  295.8   295.8  34.035 2.235e-08 ***
Age:Sex     1  227.2   227.2  26.136 7.599e-07 ***
Residuals 194 1686.3     8.7
```

```
anova(m2)
```

```
Response: Height
           Df Sum Sq Mean Sq F value    Pr(>F)
Sex         1  480.7  480.67   13.99 0.0002412 ***
Residuals 196 6733.9   34.36
```

$$F = \frac{\frac{1}{\#\text{ parameters tested}}(SSE_{reduced} - SSE_{full})}{\frac{1}{n-p}SSE_{full}} = 290.35$$

$F_{2,194}$

mean of F distribution ≈ 1

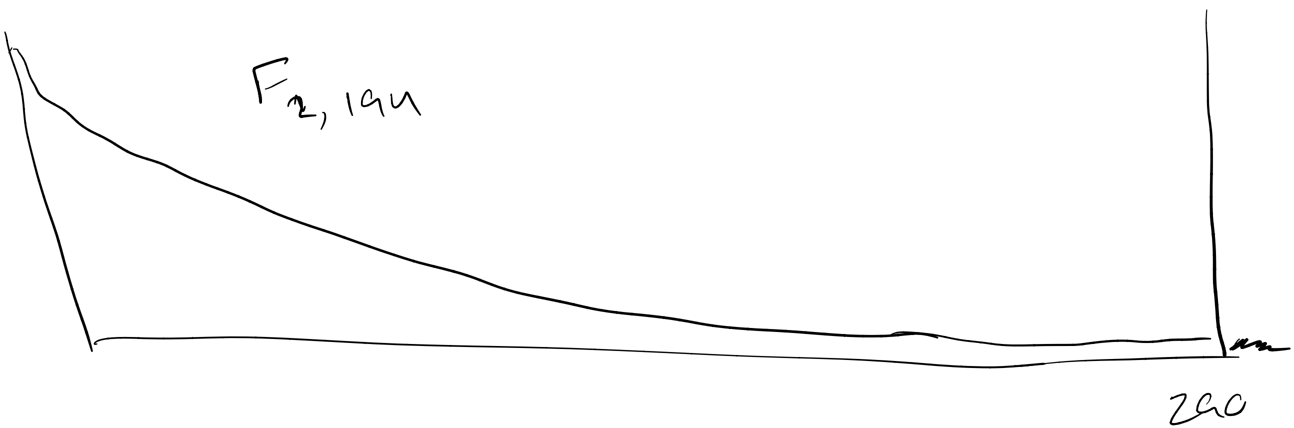
Class activity

> What is the p-value for this test?

```r
pf(290.35, df1 = 2, df2 = 194, lower.tail=F)
```

```
## [1] 4.686114e-59
```

The p-value is basically 0, so we have strong evidence that there is a relationship between Age and Height, after accounting for Sex.



Testing in R

We can do comparisons between nested models a little more efficiently:

```
m1 <- lm(Height ~ Age*Sex, data = Kids198)
m2 <- lm(Height ~ Sex, data = Kids198)
                reduced model
anova(m2, m1)
              full model
```

```
## Analysis of Variance Table
##
## Model 1: Height ~ Sex
## Model 2: Height ~ Age * Sex
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    196 6733.9
## 2    194 1686.3  2    5047.6 290.35 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$

> How did we test a single coefficient in simple linear regression?

Testing a single parameter

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> How did we test a single coefficient in simple linear regression?

With a t-test!

$$t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}}$$

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$

$$t = \frac{\widehat{\beta}_3 - 0}{SE_{\widehat{\beta}_3}}$$

Under H_0, this test statistic follows a t distribution.

> What do you think should be the degrees of freedom of the t distribution?

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$n - p$

Testing a single parameter

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$$H_0 : \beta_3 = 0 \qquad H_A : \beta_3 \neq 0$$

$$t = \frac{\widehat{\beta}_3 - 0}{SE_{\widehat{\beta}_3}}$$

Under H_0, this test statistic follows a t_{n-p} distribution.

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

```
m1 <- lm(Height ~ Age*Sex, data = Kids198)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 32.960916   1.537781  21.434  < 2e-16 ***
Age          0.182543   0.009383  19.454  < 2e-16 ***
Sex          7.727846   2.034957   3.798 0.000195 ***
Age:Sex     -0.064134   0.012545  -5.112  7.6e-07 ***
```

$H_0: \beta_3 = 0$

$H_A: \beta_3 \neq 0$

$$t = \frac{\widehat{\beta}_3 - 0}{SE_{\widehat{\beta}_3}} = \frac{-0.0641}{0.0125} = -5.112$$

p-value = 7.6×10^{-7}

Could also do a nested F test, in which case

$F = t^2 \sim F_{1,n-p}$ (exactly same p-value)

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on Age *larger* for boys than for girls?

> What hypotheses could I test to answer this research question?

$H_0: \beta_3 = 0 \qquad H_A: \beta_3 > 0$

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

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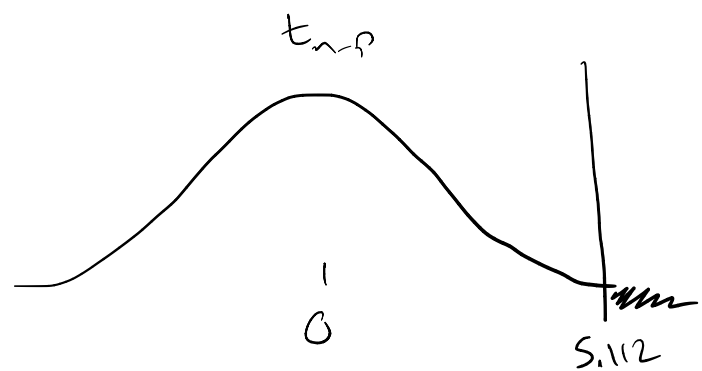
Test statistic: $t = \dfrac{\widehat{\beta}_3 - 0}{SE_{\widehat{\beta}_3}} = 5.112$



> Which tail do I use to calculate the p-value? right

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on Age *larger* for boys than for girls?

$$H_0 : \beta_3 = 0 \qquad H_A : \beta_3 > 0$$

Test statistic: $t = \dfrac{\widehat{\beta}_3 - 0}{SE_{\widehat{\beta}_3}} = 5.112$

```
pt(5.112, df = 194, lower.tail=F)
```

```
## [1] 3.805925e-07
```

p-value = 3.8×10^{-7}, so we have strong evidence that the slope on Age is larger for boys than for girls.

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on Age *larger* for boys than for girls?

$H_0 : \beta_3 = 0 \qquad H_A : \beta_3 > 0$

> Can I use a nested F test here?

No -- a nested F test doesn't let me test a specific direction ($\beta_3 > 0$)

t tests vs. nested F tests

t test: test a single parameter

Some example hypotheses:

- $$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$
- $$H_0 : \beta_3 = 0 \quad H_A : \beta_3 > 0$$
- $$H_0 : \beta_3 = 1 \quad H_A : \beta_3 < 1$$

nested F test: test one or more parameters

Some example hypotheses:

$$H_0 : \beta_1 = \beta_2 = 0$$
$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Confidence intervals

Just like with simple linear regression, we can construct confidence intervals for single parameters.

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval for β_3:

$$\widehat{\beta}_3 \pm t^* SE_{\widehat{\beta}_3}$$

where t^* is the critical value for a t_{n-p} distribution.

Confidence intervals

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval: $\widehat{\beta}_3 \pm t^* SE_{\widehat{\beta}_3}$

> Suppose I want to calculate a 99% confidence interval. How do I find the critical value t^*?

Confidence intervals

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval: $\widehat{\beta}_3 \pm t^* SE_{\widehat{\beta}_3}$

> Suppose I want to calculate a 99% confidence interval. How do I find the critical value t^*?

```
qt(0.005, df=194, lower.tail=F)
```

```
## [1] 2.601409
```

$t^* = 2.60$

Confidence intervals

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval: $\widehat{\beta}_3 \pm t^* SE_{\widehat{\beta}_3}$

$$0.0641 \pm 2.60 \cdot 0.0125 = (0.0316, 0.0966)$$

So we are 99% confident that a one-month increase in age is associated with an change in height that is between 0.0316 and 0.0966 inches greater for boys than for girls.